TUTORIAL 2
PROGRAM DESIGN

Selection with Flowcharts and Pseudocode

What is a flowchart

- Flow charting is the first step we take in understanding a process
- Flow charts provide a visual illustration, a picture of the steps the process undergoes to complete it's assigned task
- Organized combination of shapes, lines, and text

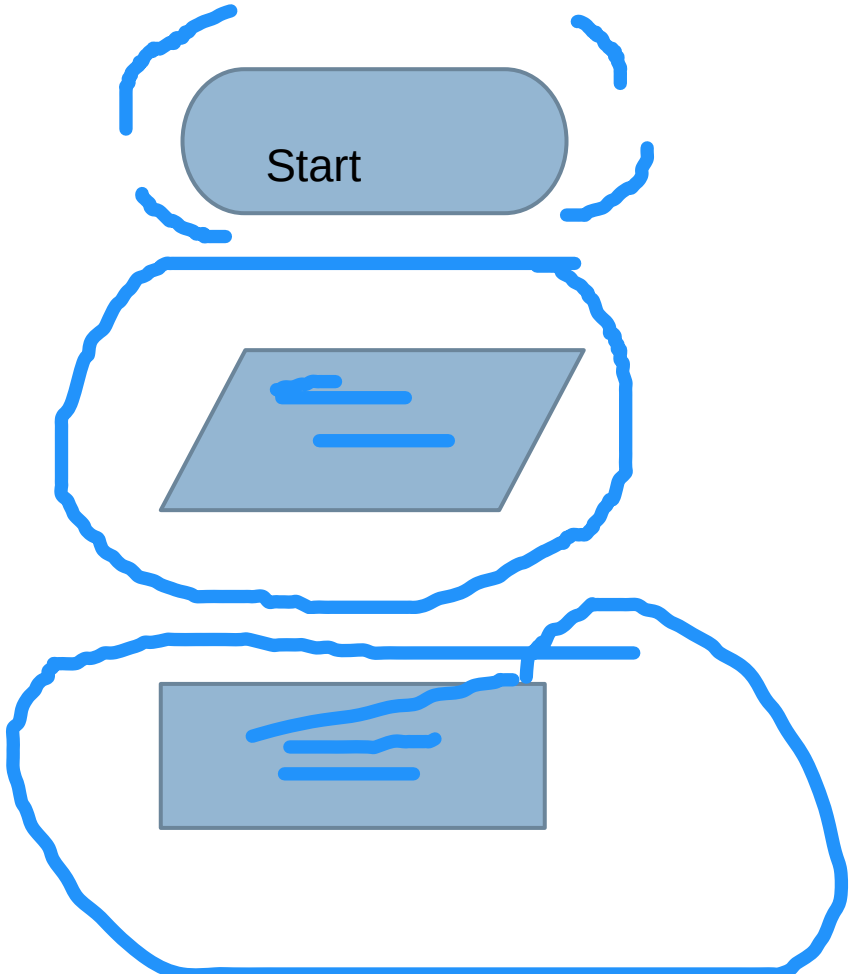
Symbols in Flowcharts

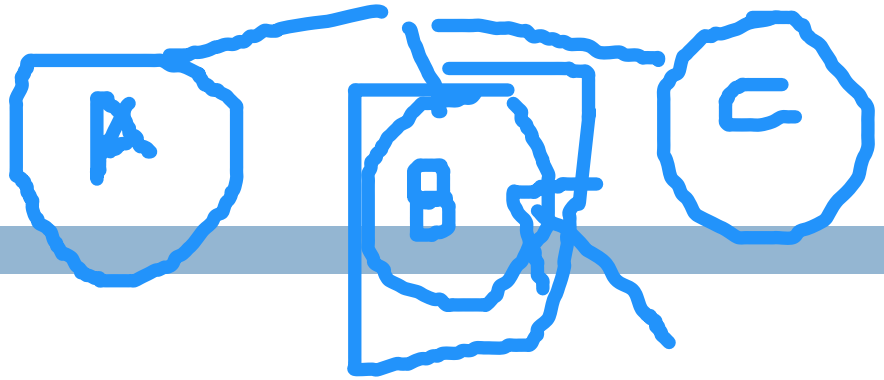
There are 6 symbols commonly used

1. Terminal :
   indicates the start / end
   of the program

2. Input/ Output
   Use for Input/Output
   (I/O) operation.

3. Process
   An internal operation
   (any processing taking place)

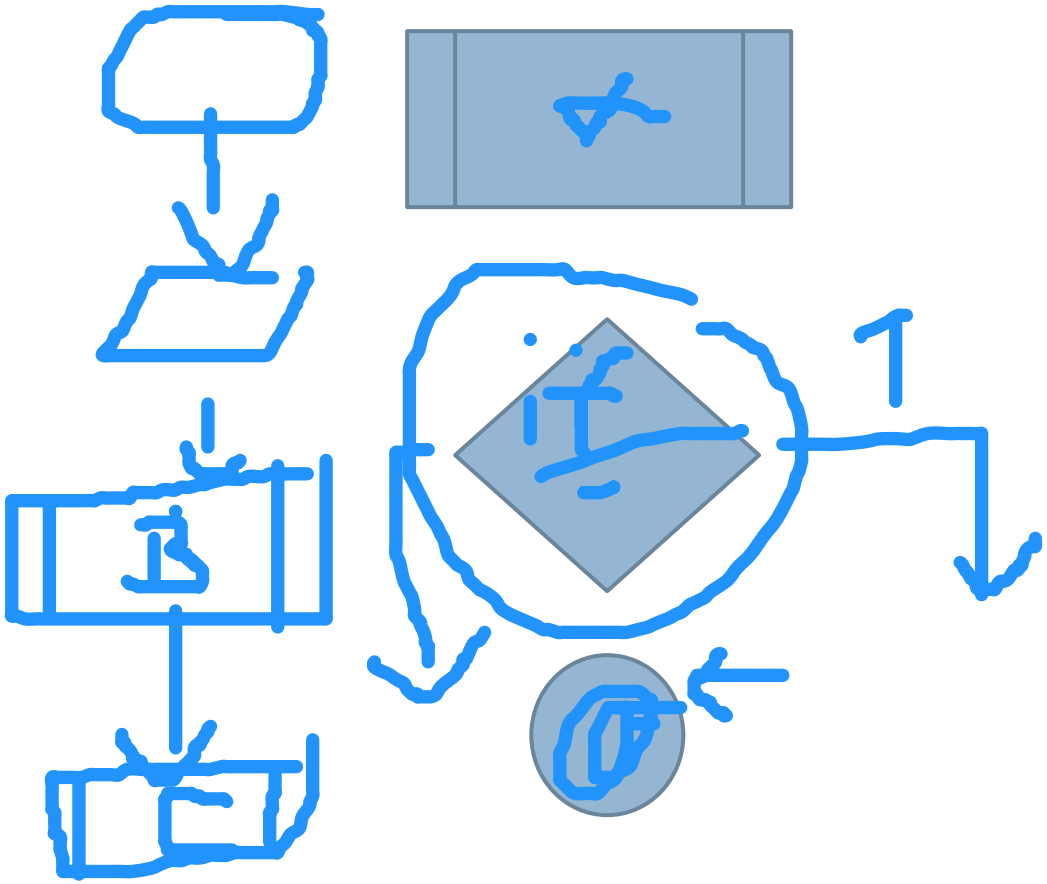


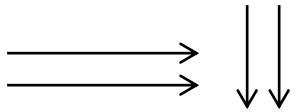


4. Pre-defined Process:
   A named process such
   as a sub-routine or a module.

5. Decision:
   Decision making
   and branching

6. Connector:
   A jump from one point
   in the flow to another

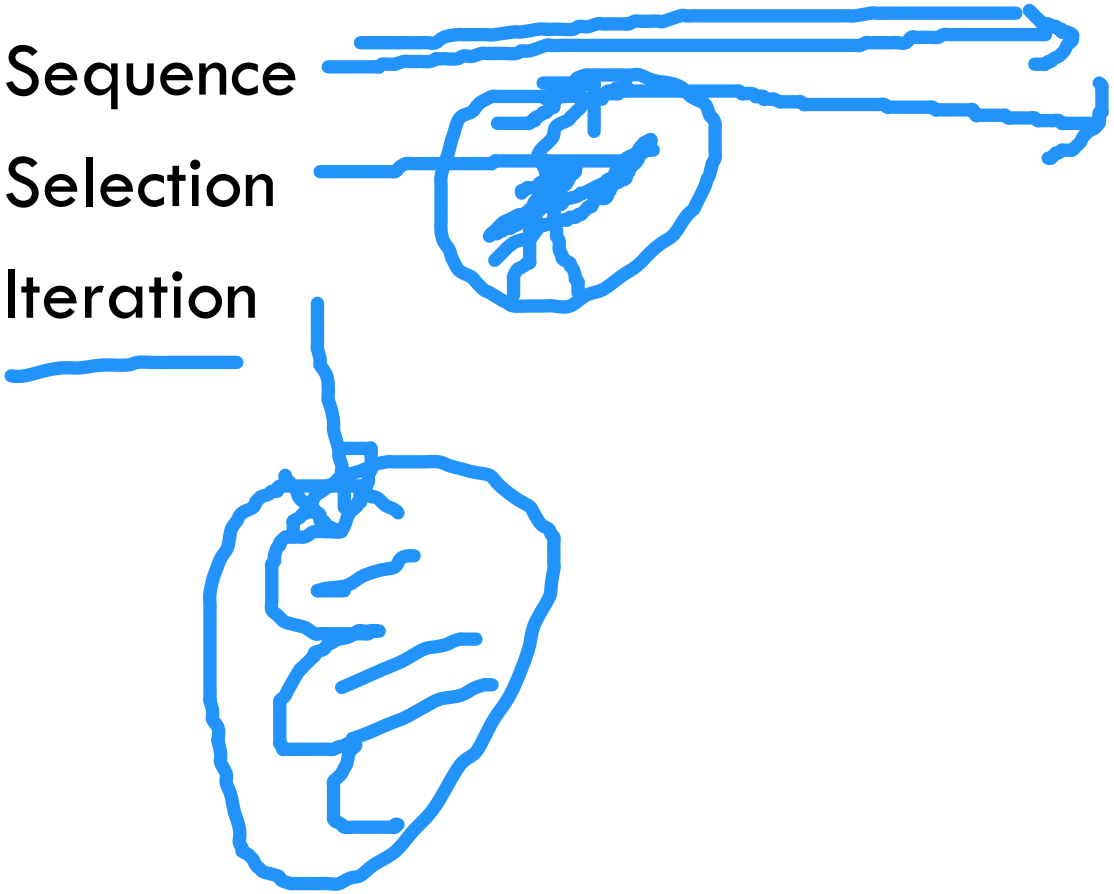




control flow

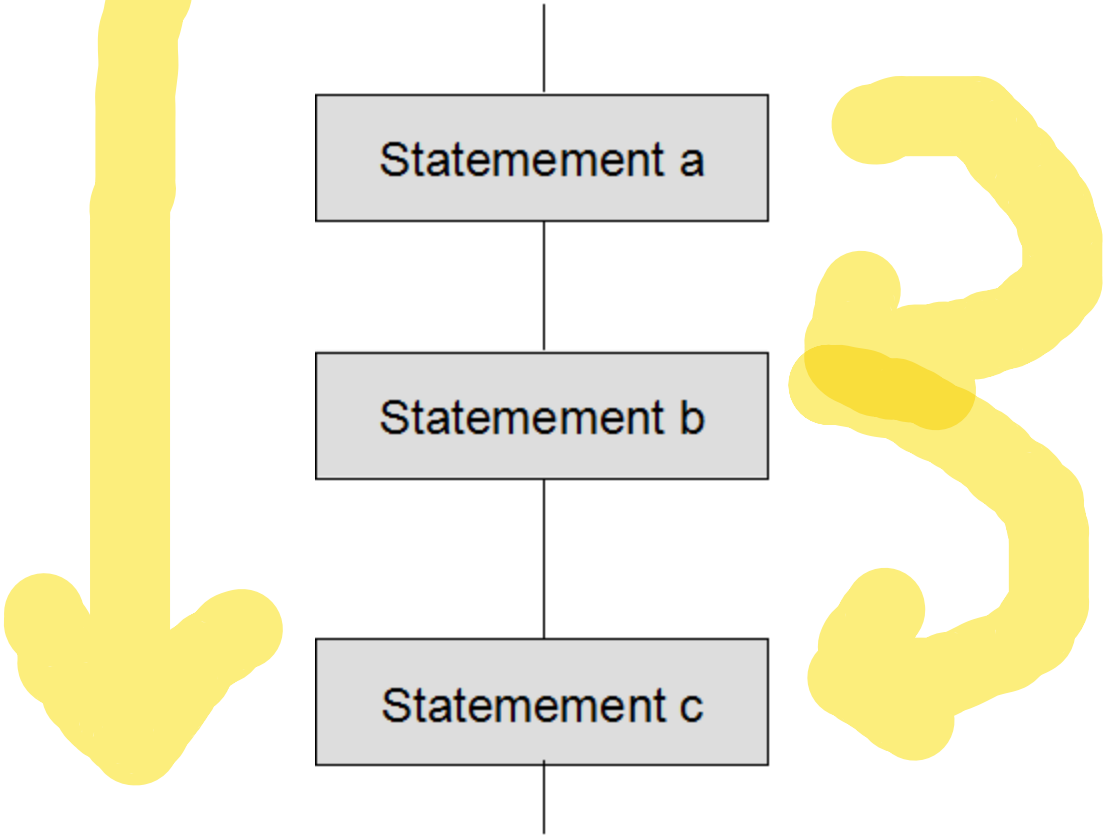
Control Structures in an Algorithm

1. Sequence
2. Selection
3. Iteration



Sequence

- sequence is a set of instructions that are in a logical order, ie. one instruction after the other



Example 1 – Calculating Years and Days

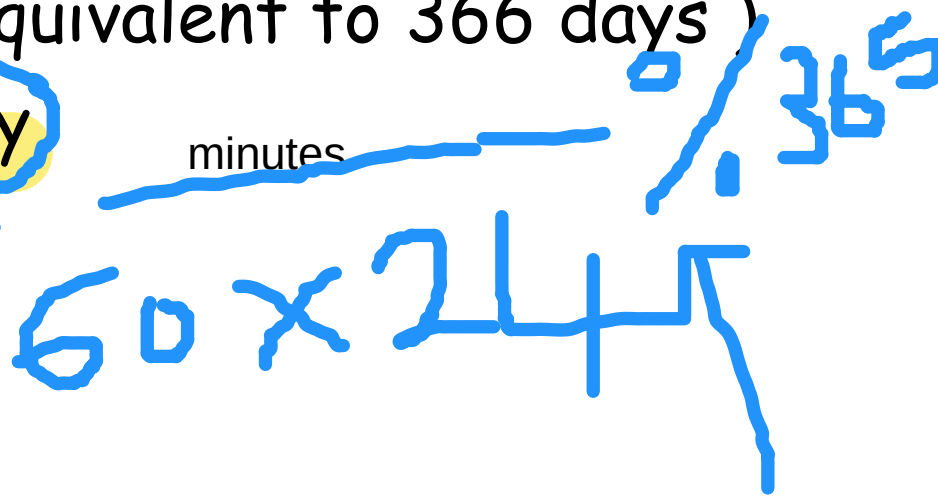
4 % 2

Write an algorithm to convert the input minutes into the number of years and days.

Eg:

Input(mint) : 527040 (equivalent to 366 days)

Ouput : 1 year 1 day



1. Read minutes from the user

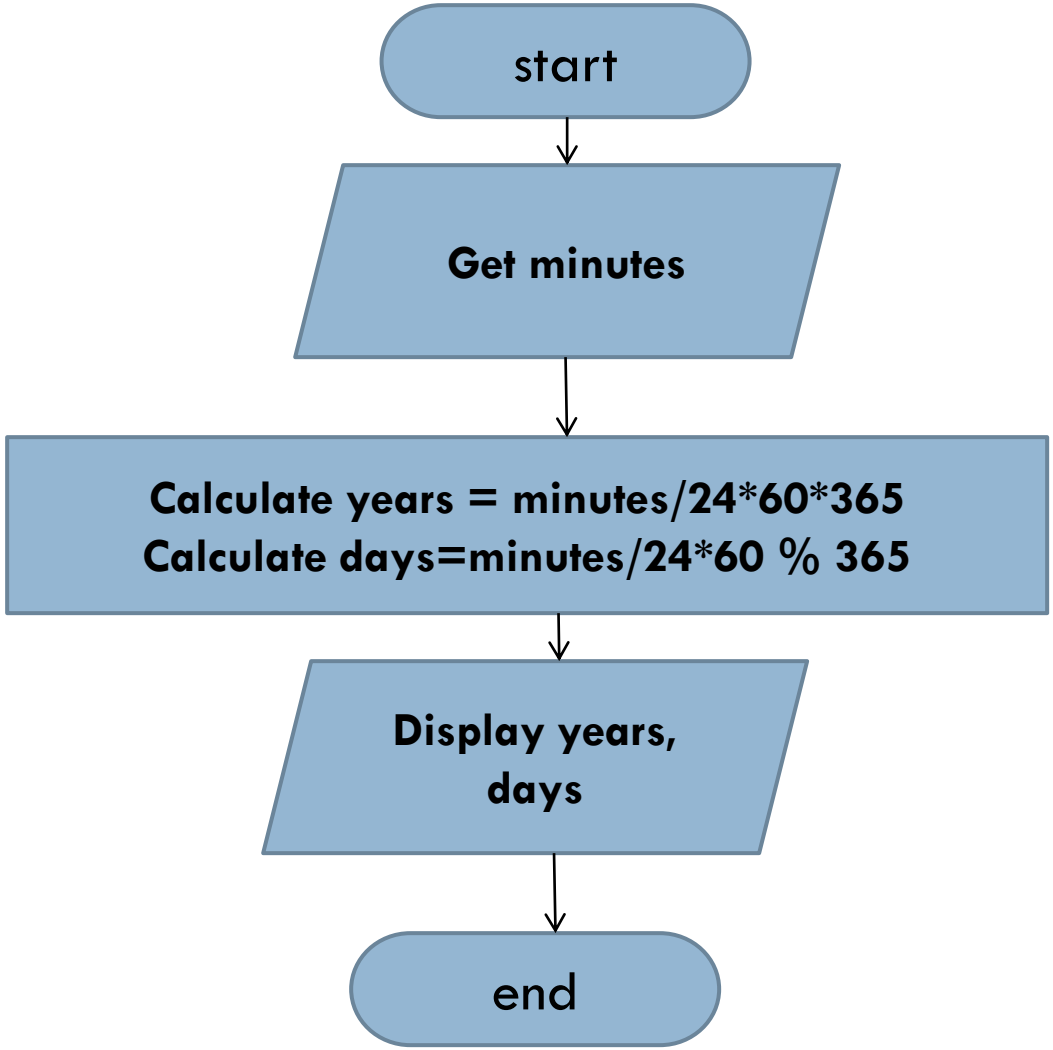
2. Calculate the years as minutes/60*24*365

3. Calculate the days as (minutes /60*24) % 365

4. Display the years, days



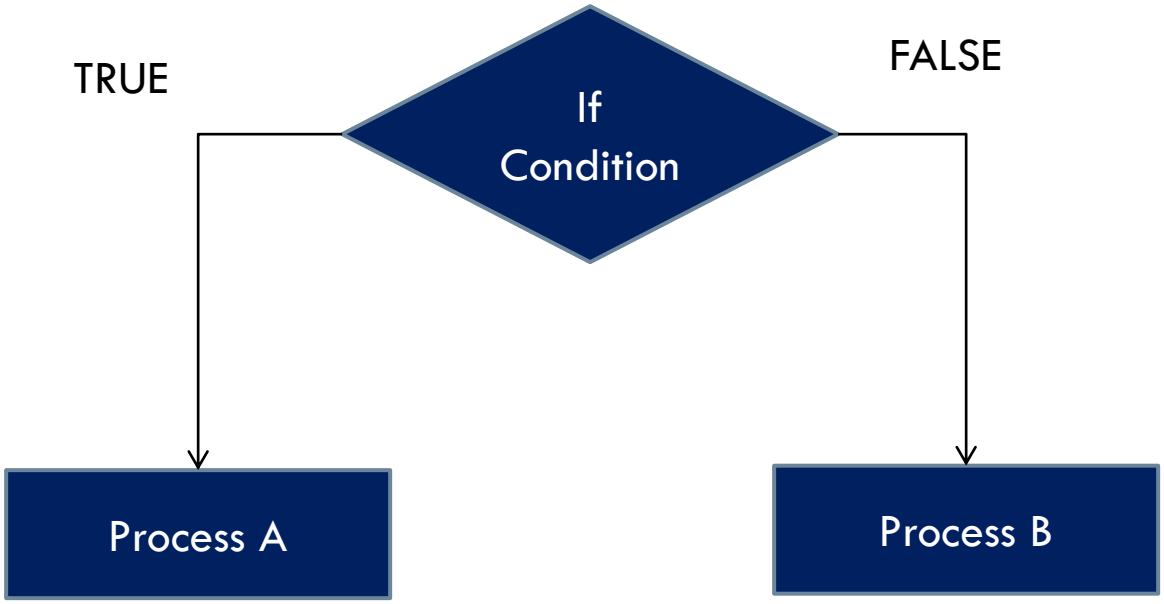
Flowchart


start
Get minutes
Calculate years = minutes/24*60*365
Calculate days=minutes/24*60 % 365
Display years,
days
end

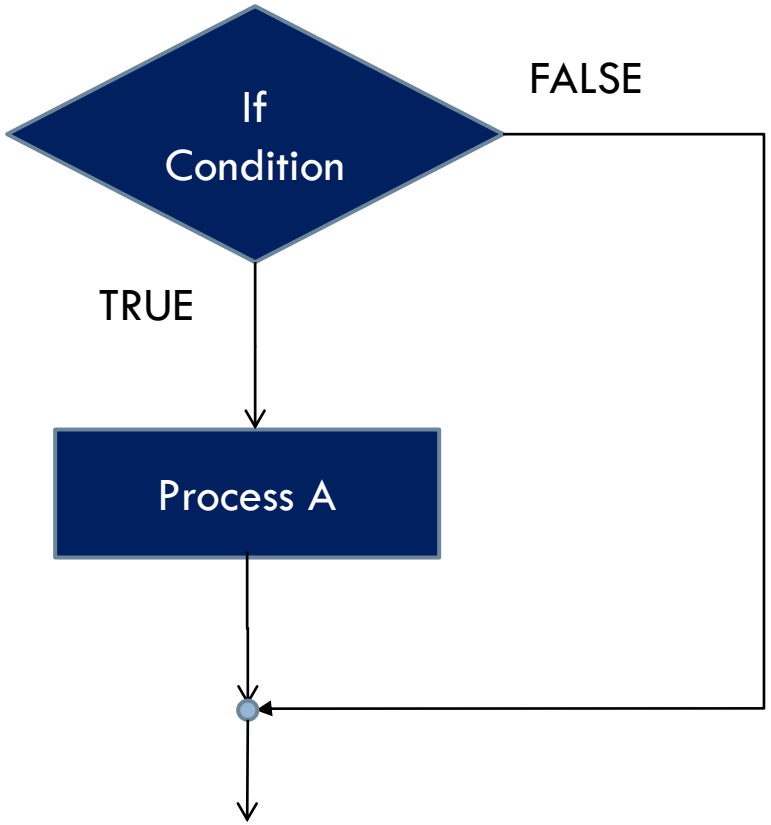
Selection

- The selection structure is where statements can be executed or skipped depending on whether a condition evaluates to TRUE or FALSE
- There are three types of selection structures
  - Simple IF – ELSE
  - IF (NULL false Condition)
  - Switch (or Case)
- Selections may be extended to
  - Nested IF
  - Combined IF Conditions

Simple IF-ELSE



IF (NULL False Condition)



Pseudocode



Eg:

```
IF std_gender = "female" THEN
        Add 1 to total_females
ELSE
        Add 1 to total_males
ENDIF
```

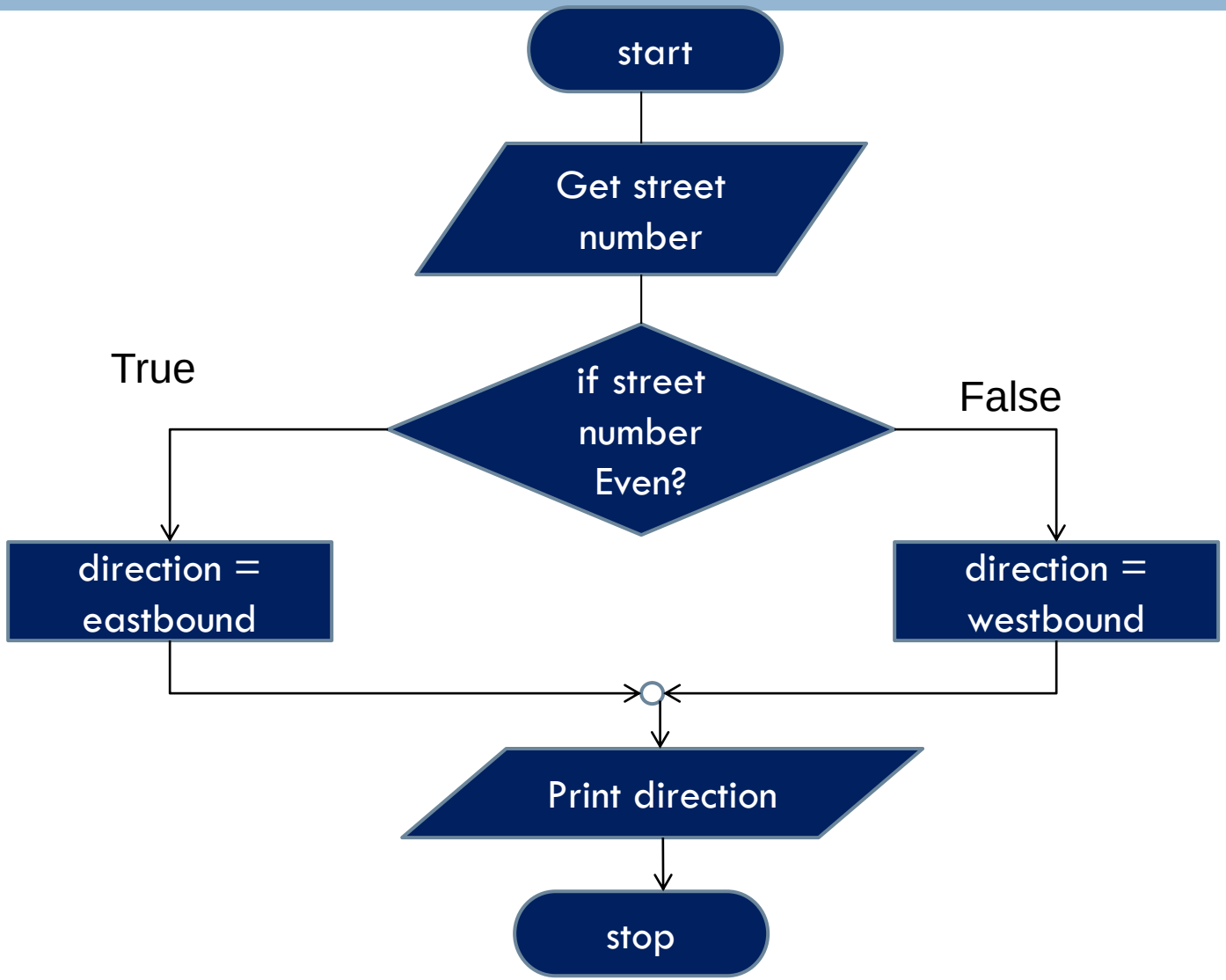
Example 2 - Direction of Numbered NYC Streets

User inputs a street number of a one-way street in New York City, print the direction of the street, either eastbound or westbound, to the screen.

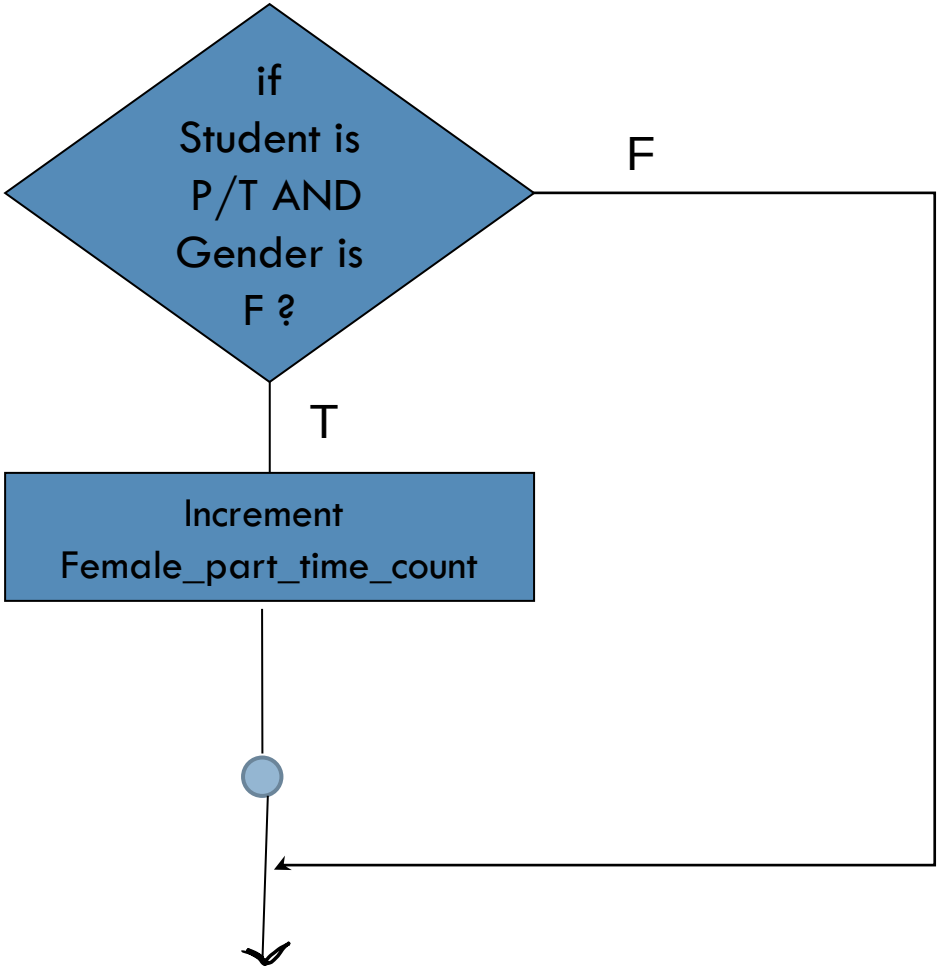
Note:

In New York City even numbered streets are Eastbound, odd numbered streets are Westbound

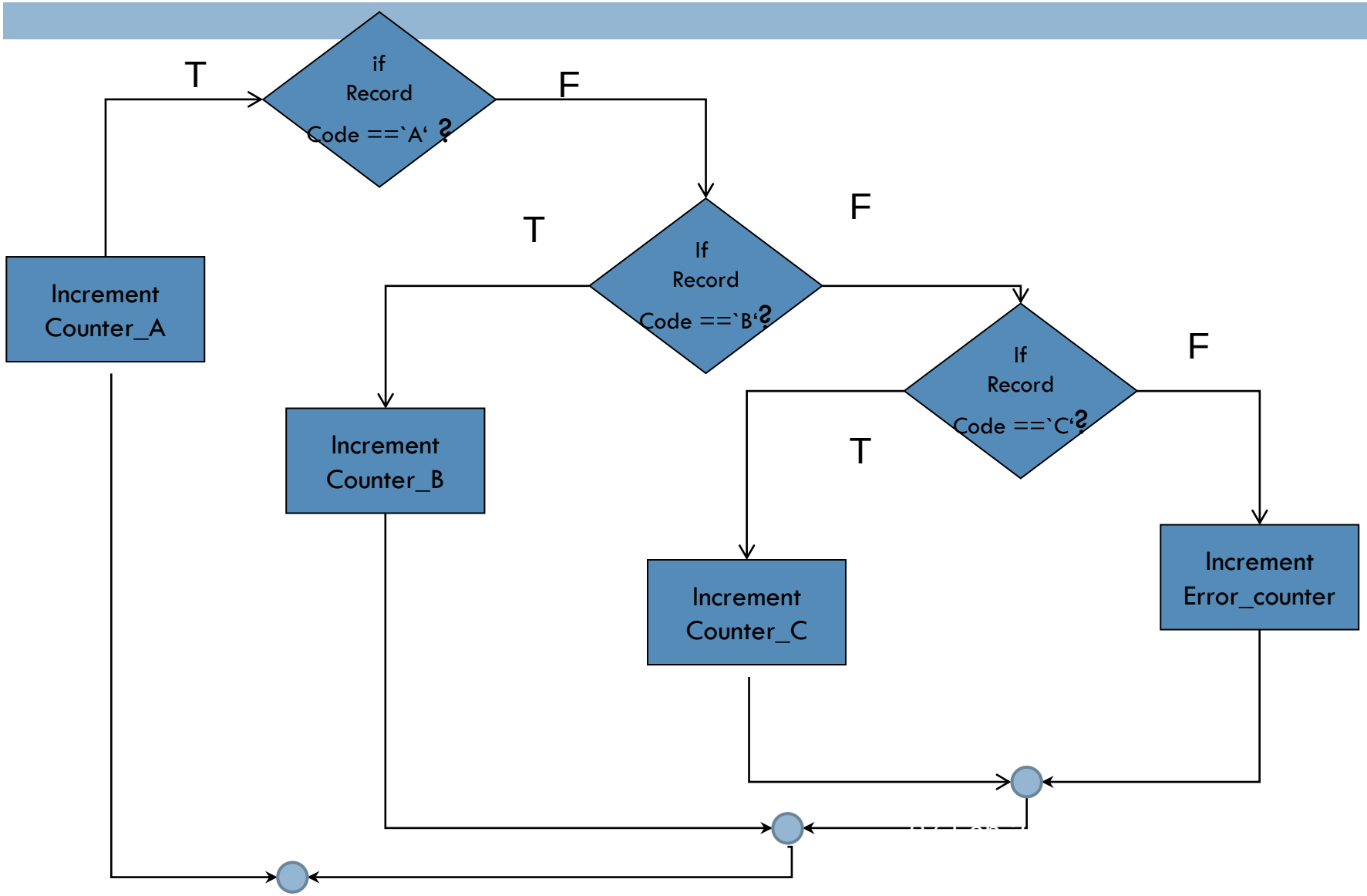
Solution


start
Get street number
if street number Even?
True
False
direction = eastbound
direction = westbound
Print direction
stop

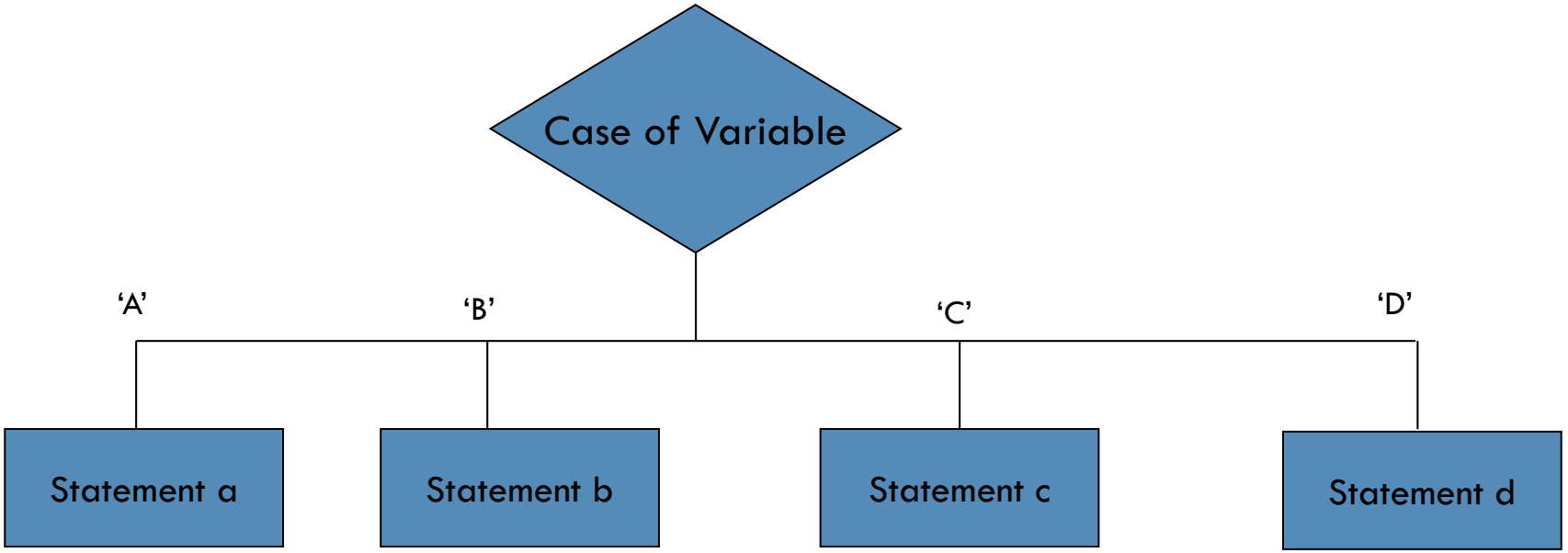
Combined IF statement



Nested IF statement



Case Structure



Example 3 – Pricing Algorithm

In a pharmacy the final price for the medicine is calculated by applying a tax on the barcoded price.

The tax percentages for the medicine categories are as follows:

Category	tax %
A	2.0
B	2.5
C	3.0

Represent the algorithm in flowchart in calculating the price for medicine, if a customer purchases a dozen of capsules where each is marked at $2.50 in category B.